



**Empowering Future
Tech Professionals**

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TASK BOOKLET – DATA SCIENCE

Program: EncoderX Remote Internship Batch 02
Track: Data Science
Week: 01

★
LEARN
BUILD
INNOVATE
SUCCEED
★★★

ABOUT ENCODERX

EncoderX provides industry-oriented remote internship experiences designed to help students build practical skills, develop professional portfolios, and gain real-world project exposure.

DO'S

- ✓ Maintain professional communication.
- ✓ Use GitHub for version control.
- ✓ Document your work clearly.
- ✓ Follow coding standards and best practices.
- ✓ Submit before deadlines.

DON'TS

- ✗ Do not submit copied projects.
- ✗ Do not share confidential project material.
- ✗ Avoid incomplete documentation.
- ✗ Do not falsify results or reports.

TASK 1: EXPLORATORY DATA ANALYSIS

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EXPECTED DURATION

1 Week

D

DIFFICULTY LEVEL

Beginner to Intermediate

T

TOOLS

Python / Power BI / Tableau

A

DOMAIN

Data Analysis

OBJECTIVE

The purpose of this task is to develop a strong understanding of data exploration, preprocessing, and analytical thinking. Interns will work with a real-world public dataset to identify patterns, trends, correlations, and actionable insights that can support data-driven decision-making.

TASK OVERVIEW

Exploratory Data Analysis (EDA) is a critical first step in any data science project. In this task, interns are required to select a publicly available dataset from platforms such as Kaggle, UCI Machine Learning Repository, Data.gov, or other reliable sources and perform a comprehensive analysis. The goal is not only to understand the dataset but also to communicate meaningful findings through visualizations and reports.

LEARNING OUTCOMES

Upon successful completion of this task, interns will be able to:

- ✓ Understand the structure and characteristics of a dataset.
- ✓ Perform data cleaning and preprocessing techniques.
- ✓ Handle missing values and outliers effectively.
- ✓ Create informative visualizations using industry-standard tools.
- ✓ Identify relationships and trends within data.
- ✓ Present findings in a professional and structured manner.

INSTRUCTIONS

1**Step 1: Dataset Selection**

Choose a publicly available dataset containing at least 1,000 records and multiple features – e.g. Sales and Revenue Data, Healthcare Records, Customer Behavior Data, E-commerce Transactions, Education Statistics, or Social Media Analytics.

2**Step 2: Data Cleaning**

Remove duplicate records; handle missing values appropriately; correct data inconsistencies; convert data types where necessary; detect and treat outliers.

3**Step 3: Exploratory Analysis**

Analyze the dataset by identifying key trends, the strongest relationships between variables, any unusual patterns or anomalies, and the practical insights that can be derived.

4**Step 4: Data Visualization**

Create at least five professional visualizations – Bar Charts, Line Charts, Histograms, Correlation Heatmaps, and Scatter Plots – using tools such as Python (Matplotlib, Seaborn, Plotly), Power BI, or Tableau.

5**Step 5: Insights & Recommendations**

Summarize your findings and provide actionable recommendations based on the analysis.

DELIVERABLES

- ✓ Complete Jupyter Notebook (.ipynb)
- ✓ Cleaned Dataset (if applicable)
- ✓ Professional Analysis Report (PDF)
- ✓ GitHub Repository Link
- ✓ Screenshots of Visualizations
- ✓ README.md Documentation

EVALUATION CRITERIA

Criteria	Weightage
Data Cleaning & Preparation	20%
Quality of Analysis	25%
Data Visualizations	20%
Insights & Recommendations	20%
Documentation & GitHub Repository	15%

SUBMISSION REQUIREMENTS

- ✓ Upload all project files to a public GitHub repository.
- ✓ Maintain a clear and professional README file.
- ✓ Record a 3–5 minute project demonstration video.
- ✓ Publish a LinkedIn post describing your project and key findings.
- ✓ Tag EncoderX in the post.
- ✓ Include hashtags: [#EncoderX](#) [#DataScience](#) [#Internship](#) [#LearningInPublic](#)
- ✓ Submit the GitHub Repository Link, LinkedIn Post Link, and Video Link in a single PDF through the assigned submission portal.

● ————— ALL THE BEST! ————— ●